

ALFA RIZI

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Early-career Software Engineer and Informatics Management student (GPA 3.6/4.0) with practical experience building distributed microservices, RAG AI pipelines, and responsive web applications. Strong technical foundation in Java (Spring Boot 3, Spring Security, Spring Cloud Gateway), TypeScript/JavaScript (React, Next.js, Express), Python (FastAPI), and PostgreSQL (PL/pgSQL triggers, views, Flyway). Experienced with Docker containerization and automated GitHub Actions CI/CD workflows.

TECHNICAL SKILLS

Languages:	Java 17, TypeScript, JavaScript (ES6+), Python 3.12, SQL, C
Backend & Microservices:	Spring Boot 3.2, Spring Cloud Gateway, Spring Security 6, Spring Data JPA, FastAPI, Express.js
Frontend Development:	React 18/19, Next.js 16 (App Router), HTML5, CSS3, Tailwind CSS, Canvas API, Web Speech API
Databases & ORM:	PostgreSQL (PL/pgSQL Triggers, Views), Drizzle ORM, Flyway, Hibernate, ChromaDB, SQLite
DevOps & Quality:	Docker, Docker Compose, GitHub Actions (CI/CD), Git, Maven, Postman, Linux CLI, OpenAPI/Swagger

FEATURED ENGINEERING PROJECTS

ApexGrid — F1 Race Ticket Booking Platform | *Java 17, Spring Boot, React, PostgreSQL, Docker* 2026

- Architected a 3-service distributed system (API Gateway, User Service, Race Service) using Spring Boot 3.2 and Spring Cloud Gateway.
- Implemented stateless JWT authentication with refresh token rotation and RBAC (USER/ADMIN) enforced across service levels.
- Built real-time quota validation using PostgreSQL PL/pgSQL triggers and automated 30-minute booking cancellations via Spring @Scheduled.
- Developed an admin dashboard and race booking UI in React 18 with Tailwind CSS; containerized the complete stack using Docker Compose.

StudyMate AI — Asynchronous RAG Document Assistant | *Python, FastAPI, ChromaDB, Sentence-Transformers* 2026

- Engineered an asynchronous RAG pipeline using FastAPI, PyMuPDF, and LangChain splitters to index PDF textbooks with verified page citations.
- Implemented semantic search via Sentence Transformers (all-MiniLM-L6-v2) embeddings indexed into persistent ChromaDB collections.
- Orchestrated prompt pipelines with an LLM gateway to generate structured summaries, multi-choice quizzes, and interactive flashcards.
- Configured automated CI/CD via GitHub Actions executing Ruff linting, Pytest async suites, and pip-audit vulnerability checks.

Finance Feasibility — AI-Assisted Investment Analysis | *React 19, Express.js, PostgreSQL, Drizzle ORM, Gemini* 2026

Demo: <https://finance-feasibility.alfarizi.my.id> | Code: github.com/a1fariz/finance-feasibility

- Developed an investment appraisal application calculating DCF, NPV, IRR, ROI, and Payback Period with automated Gemini AI evaluation.
- Built a type-safe REST API in Express.js with PostgreSQL and Drizzle ORM, secured via Firebase Admin auth and rate limiting.
- Constructed interactive financial visualization charts in React 19 using Recharts and implemented client-side Excel (XLSX) report export.

Renshuu — Interactive Japanese Learning Platform | *Next.js 16, React 19, TypeScript, Canvas API, Web Speech API* 2026

Demo: <https://renshuu.alfarizi.my.id> | Code: github.com/a1fariz/renshuu

- Built a language learning platform covering Kana, Kanji (KanjiVG SVG stroke data), Vocabulary, and JLPT N4 Grammar.
- Created an interactive HTML5 Canvas character tracing pad and integrated native browser Web Speech API (ja-JP) for audio playback.
- Implemented a Spaced Repetition (SRS) review queue and offline-first LocalStorage state management with JSON backup/restore.

PROFESSIONAL EXPERIENCE

Lead Intern — IT & Business Operations 2025 – Present

PT Resik Cemerlang (Pasim Group)

Bandung, Indonesia

- Directed a 5-person intern team managing day-to-day IT support operations, hardware diagnostics, and workstation availability.
- Enforced SLA ticket resolution, executing security patching, software configurations, and antivirus definition updates across laptops.
- Conducted quality control and data validation across thousands of medical device specification records to ensure database accuracy.
- Prepared structured weekly operational health and incident summary reports for senior management review.

Lead Intern — IT Hardware 2024

SMP Negeri 2 Batujajar

West Bandung, Indonesia

- Diagnosed and resolved hardware faults, network cabling errors, and printer issues across school computer laboratories.
- Established a physical hardware inventory and maintenance tracking log that reduced repeat-fault turnaround time.

EDUCATION & TRAINING

Universitas Nasional PASIM Expected May 2027

Associate Degree in Informatics Management (D3) — GPA: 3.6 / 4.0

Bandung, Indonesia

- PUB Intensive Technical Scholarship (2024 – Present): Logic & C, Data Structures, Relational Databases, Advanced Java Development.